LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: The Human Eye and the Colourful World

Parts of the eye

Part	Job
Cornea	Transparent front bulge; most of the refraction happens here
Iris	Coloured muscle that controls the size of the pupil
Pupil	Opening that controls how much light enters (smaller in bright light)
Eye lens	Convex lens that fine-tunes focus; its focal length is changed by the ciliary muscles
Retina	Light-sensitive screen where a real, inverted image forms. Signals go to the brain through the optic nerve

- **Accommodation:** the eye lens changes its focal length (thin for distant objects, thick for near ones).
- Normal eye: near point 25 cm (least distance of distinct vision); far point at infinity.

Defects of vision

Defect	Can't see clearly	Cause	Correction
Myopia (near-sightedness)	Distant objects (image forms in front of the retina)	Lens too curved / eyeball too long	Concave lens
Hypermetropia (far-sightedness)	Nearby objects (image forms behind the retina)	Focal length too long / eyeball too short	Convex lens
Presbyopia	Nearby objects, with age	Ciliary muscles weaken; lens loses flexibility	Convex lens, or bifocals

Prism & dispersion

- A glass prism splits white light into a **spectrum**: VIBGYOR. **Red bends the least, violet the most.**
- Newton: a second, identical prism turned upside down recombines the colours into white light.
- **Rainbow:** forms opposite the Sun. Water droplets act as prisms: refraction + dispersion → internal reflection → refraction.

Atmospheric refraction

- **Twinkling of stars:** starlight passes through air layers with changing refractive index, so the star's position and brightness flicker. **Planets don't twinkle**: they are closer and act as extended sources, so the variations average out.
- **Advance sunrise / delayed sunset:** the Sun is seen about 2 minutes before actual sunrise and about 2 minutes after actual sunset.

Scattering of light

- **Tyndall effect:** colloidal particles scatter light, making a beam visible (sunlight through trees or a dusty room).
- **Blue sky:** fine particles scatter blue (short wavelength) light much more than red.
- **Red sunrise and sunset:** light travels a long path near the horizon, blue is scattered away, and red reaches us.
- **Danger signals are red:** red is scattered least by fog and smoke, so it is visible from far.

- Clouds look white: larger particles scatter all colours about equally. In space the sky would look dark (no scattering).

Quick check

1. Which lens corrects myopia, and why?

2. Why do planets not twinkle?

3. Why does the sky appear dark to an astronaut in space?